A Two-Dimensional Dynamic Anatomical Model of the Human Knee Joint

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The objective of this study is to develop a two-dimensional dynamic model of the knee joint to simulate its response under sudden impact. The knee joint is modeled as two rigid bodies, representing a fixed femur and a moving tibia, connected by 10 nonlinear springs representing the different fibers of the anterior and posterior cruciate ligaments, the medial and lateral collateral ligaments, and the posterior part of the capsule. In the analysis, the joint profiles were represented by polynomials. Model equations include three nonlinear differential equations of motion and three nonlinear algebraic equations representing the geometric constraints. A single point contact was assumed to exist at all times. Numerical solutions were obtained by applying Newmark constant-average-acceleration scheme of differential approximation to transform the motion equations into a set of nonlinear simultaneous algebraic equations. The equations reduced thus to six nonlinear algebraic equations in six unknowns. The Newton-Raphson iteration technique was then used to obtain the solution. Knee response was determined under sudden rectangular pulsing posterior forces applied to the tibia and having different amplitudes and durations. The results indicate that increasing pulse amplitude and/or duration produced a decrease in the magnitude of the tibio-femoral contact force, indicating thus a reduction in the joint stiffness. It was found that the anterior fibers of the posterior cruciate and the medial collateral ligaments are the primary restraints for a posterior forcing pulse in the range of 20 to 90 degrees of knee flexion; this explains why most isolated posterior cruciate ligament injuries and combined injuries to the posterior cruciate ligament and the medial collateral results from a posterior impact on a flexed knee.

Introduction

Historically, research on the mechanics of the human musculoskeletal system has focused on experimental studies. Recently mathematical tools from the field of applied mechanics have been used to obtain a better understanding of the complicated mechanical behavior of the substructures which comprise the knee.

In a recent review of knee models, Hefzy and Grood [1] classified knee models into phenomenological and anatomical based models. The phenomenological models have limited usage since they describe the response of the knee without considering its real structures. In a sense, these models are not really "knee models" since their effectiveness in accurately predicting in vivo response depends heavily on the proper simulation of its articulating surfaces. On the other hand, the anatomical models have been developed to study the behavior of various structural components forming the knee joint and thus require accurate geometric description of the real anatomy.

Knee joint models can be divided into the following four categories [1]: (*a*) models of load sharing, (*b*) models of ligament length, (*c*) models of femoral-tibial contact stresses, and (*d*) models including ligamentous structure and geometric constraints. The model presented in this paper falls under the fourth category. In this category the three-dimensional models developed by Andriacchi et al. [2], Wismans et al. [3, 4], Essinger et al. [5], and Blankevoort and Huiskes [6] are the most comprehensive anatomical quasi-static models available in literature, and the two-dimensional models developed by Moeinzadeh et al. [7–11] and Wongchaisuwat et al. [12] are the only anatomical dynamic models available in literature.

Moeinzadeh et al.'s [7–12] presented a two-dimensional dynamic model of the knee joint. The femur and the tibia were modeled as rigid bodies with the femur being fixed and the tibia undergoing planar motion in the sagittal plane. Four ligaments, the two cruciates and the two collaterals, modeled as nonlinear elastic elements, were considered. All ligaments would carry a force only if their current lengths were longer than their initial lengths which were determined when the tibia was positioned at 54.79 degrees of knee flexion. A one contact point analysis was conducted where normals to the surfaces of the femur and the tibia at the point of contact were collinear. The profiles of the femoral and tibial articular surfaces were measured from X-rays using a two-dimensional sonic digitizing technique. A polynomial equation was used as an approximate

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mathematical representation of the profile of the surface under consideration. Results were presented only for motion from 54.79 degrees of knee flexion to full extension.

Moeinzadeh, et al.'s theoretical formulation [7–11] included three differential equations describing the motion of the tibia with respect to the femur, and three algebraic equations describing the contact condition and the geometric compatibility of the problem. Using Newmark's constant-average-acceleration scheme [13], the three differential equations of motion were transformed to three nonlinear algebraic equations. The system thus reduced to six nonlinear equations in six independent unknowns: the x and y coordinates of the origin of the tibial coordinate system with respect to the femoral one, the angle of the knee flexion, the magnitude of the contact force, and the x coordinates of the contact point in both femoral and tibial systems of axes. However, in their numerical analysis, instead of using the differential form of the Newton-Raphson iteration method to solve these six nonlinear algebraic equations, they reformulated the system to include 22 equations in 22 unknowns. This system was then solved iteratively using an incremental form of the Newton-Raphson iteration technique. In this formulation they considered the coordinates of the ligamentous tibial insertion sites (moving points) as 8 independent variables and added 8 compatibility equations for the locations of these ligamentous tibial insertion sites. The remaining independent variables included: (i) the x and y components of the unit vectors normal to the femoral and tibial profiles at the point of contact (4 variables), (ii) the y coordinates of the contact point in both the femoral and tibial coordinate systems (2 variables), (iii) the slope of the articular profiles at the contact point expressed in both femoral and tibial coordinate systems (2 variables). Numerical results were obtained for forcing functions passing through the tibial center of mass. No external moments were considered in the numerical calculation.

Wongchaisuwat et al. [12] presented a completely different dynamic model to analyze the planar motion between the femoral and tibial contact surfaces in the sagittal plane. In their model, the authors considered the tibia as a pendulum that swings about the femur. Newton-Euler equations were then used to formulate the gliding and rolling motions which were defined by holonomic and nonholonomic conditions, respectively. Using their model, the authors presented a control strategy to cause the motion and maintain the contact between the surfaces. Their control system included two classes of input: muscle forces which stabilized and caused the motion, and ligament forces which maintained the constraints.

Since Wongchaisuwat et al.'s model [12] is only a control strategy to cause and maintain contact between the femur and the tibia, the only real dynamic model is Moeinzadeh et al.'s model [7–11]. However, Moeinzadeh et al.'s model [7–11] was limited since it was valid only in the region of 0 to 55 degrees of knee flexion. This limitation was imposed by their geometrical representation of the femoral profile which diverged significantly from anatomical in the posterior part of the femur, and their assumption of all ligaments being just taut at 54.79 degrees of knee flexion. Furthermore Moeinzadeh, et al.'s [7–11] formulation consisted of using 22 independent variables, and solving 22 equations in 22 unknowns, thus making the solution time consuming and expensive. Expanding this formulation into three dimensions was thus prohibitively complex. Also, Moeinzadeh et al.'s perturbation of the coordinates of the ligaments causes the model to become more complicated when more ligaments are introduced, which increases the dimension of the stiffness matrix making the solution more costly. Using only four ligamentous structures, Moeinzadeh et al. (1988) were not able to solve the three-dimensional model since their solution technique required them to consider an additional 85 variables as independent, and to add 85 compatibility equations to solve a system of 101 equations in 101 unknowns.

Recently, Engin and Tumer [14] also presented two different methods to solve this problem which included three differential equations (equations of motion) and three algebraic equations (constraints). In one method, the Method of Minimal Differential Equations, the system was reduced to two differential equations by satisfying the constraint equations as well as their derivatives. The resulting two nonlinear differential equations were then solved via numerical integration. In the other method, Method of Excess Differential Equations (EDE), the algebraic constraints were converted to differential equations via second derivatives. The resulting six differential equations were then integrated numerically.

In this paper, we present a solution for the two dimensional dynamic model of the knee joint that is different than that of Moeinzadeh et al.'s [7–11] by reducing the problem to solving a system of six nonlinear algebraic equations in six independent unknowns. In the analysis, the reverse of the EDE method is performed by transforming the three differential equations of motion into three algebraic equations using Newmark method. A differential form of the Newton-Raphson method is then used to solve these six algebraic nonlinear equations. This method basically consists of differentiating the system of equations with respect to the six independent variables, solving thus six linear equations in six unknowns. Calculations are carried out by applying a rectangular forcing pulse to the tibia at its center of mass, and perpendicular to its axis. The numerical results are finally compared with the experimental data available in the literature.

Model Description

Coordinate Systems of Axes. The knee joint is modeled as two rigid bodies, the femur and the tibia, undergoing general planar motion in the sagittal plane. The femur has been assumed fixed while the tibia slides and rolls over it, without losing contact. Body coordinate systems were attached to the fixed femur (x, y) and to the moving tibia (x', y'). The origin of the femoral system was located at the intercondylar notch with the x-axis directed anteriorly and the y-axis directed distally. The origin of the tibial system was located at the center of mass of the tibia with the x-axis directed posteriorly and the y-axis directed proximally. Using these coordinate systems the position of the tibia can be completely described in the fixed femoral coordinate system, as shown in Fig. 1, in terms of the femoral coordinates of the origin of the tibial system (x_0, y_0) and the orientation of the tibial system with respect to the femoral system α. Accordingly, for an arbitrary point P located on the tibia we can write:

$$\mathbf{R}_p = \mathbf{R}_0 + [T]\mathbf{R}_p' \tag{1a}$$

where, $\mathbf{R}_p$ is the position vector of point P in the femoral coordinate system and $\mathbf{R}_p'$ is the position vector of point P in the tibial coordinate system, $\mathbf{R}_0$ is the position vector of the origin of the tibial coordinate system with respect to the femoral coordinate system, and is given by:

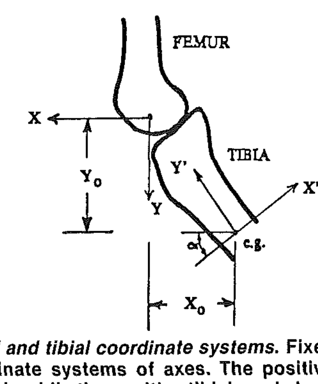

Fig. 1 *Femoral and tibial coordinate systems.* Fixed femoral and moving tibial coordinate systems of axes. The positive femoral x-axis is directed anteriorly while the positive tibial x-axis is directed posteriorly.

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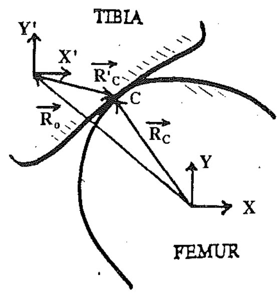

Fig. 2(*a*) *Contact condition.* This figure illustrates the contact condition: $R_c = R_0 + [T]R_c'$ where R_c and R_c' are the position vectors of the contact point in the femoral and tibial coordinate systems, respectively; R_0 is the position vector of the tibial origin with respect to the femoral origin, and $[T]$ is the rotation matrix describing the orientation of the tibia with respect to the femur.

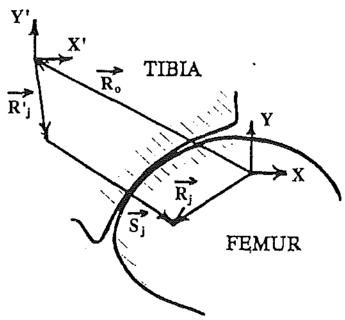

Fig. 2(*b*) *Ligamentous forces.* This figure illustrates how the direction of a ligamentous force is obtained using the relation: $S_j = R_j - R_0 - [T]R_j'$ where S_j is a vector parallel to the line of action of the force in a ligament having R_j and R_j' as the position vectors of its femoral and tibial attachment sites, respectively.

$$\mathbf{R}_0 = x_0\hat{i} + y_0\hat{j} \tag{1b}$$

and $[T]$ is the rotational transformation matrix from the tibial coordinate system to the femoral coordinate system, and is written as:

$$[T] = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \tag{1c}$$

Geometry of the Articular Surfaces. The tibial surface was modeled using the same parabolic profile employed by Moeinzadeh et al. [7–11] and determined by digitizing a radiograph of the medial condyle. The equation of this profile is given as:

$$f_2(x') = (0.2134m) - 0.0456x' + (1.073/m)x'^2 \tag{2a}$$

where x' is expressed in meters. However the femoral profile utilized was different than Moeinzadeh et al's femoral profile. Using the data reported in Zoghi et al. [15], this profile was taken as a circular arc whose equation is given by:

$$f_1(x) = k + \sqrt{R^2 - (x-h)^2} \tag{2b}$$

where;

$h = -0.0136$ m, $k = -0.0124$ m, and $R = 0.0189$ m.

This is consistent with the data reported by Walker et al. [16] who found that the posterior portions of the medial and lateral condyles are parts of spherical surfaces having mean radii of 0.02 meters.

Contact Condition. The femur and the tibia were assumed to be in contact at a point C as shown in Fig. 2(*a*). The position vector of the contact point is expressed in terms of both femoral and tibial coordinate systems as:

$$\mathbf{R}_c = x_c\hat{i} + y_c\hat{j} \quad \mathbf{R}_c' = x'_c\hat{i}' + y_c'\hat{j}' \tag{3a, b}$$

where (x_c, y_c) and (x_c', y_c') are the x and y coordinates of the contact point in the femoral and tibial systems, respectively. Since contact occurs at a point identifiable in both femoral and tibial profiles, we can write:

$$y_c = f_1(x_c) \quad y_c' = f_2(x_c') \tag{4a}$$

where $f_1(x_c)$ and $f_2(x_c')$ are given by Eqs. 2(*a*) and 2(*b*). Using Eqs. (1) the position vectors of the contact point in the femoral and the tibial coordinate systems are related by the following vectorial relation:

$$\mathbf{R}_c = \mathbf{R}_0 + [T]\mathbf{R}_c' \tag{5}$$

Equation (5) is rewritten as the following two scalar equations:

$$x_c = x_0 + R_{11}x_c' + R_{12}f_2(x_c') \tag{6}$$

$$f_1(x_c) = y_0 + R_{21}x_c' + R_{22}f_2(x_c') \tag{7}$$

where R_{ij} is the ijth component of the rotational transformation matrix, $[T]$. Satisfying Eqs. (6) and (7) at any point will insure that it is a contact point. Thus Eqs. (6) and (7) are the contact condition.

Geometric Compatibility Condition. To exclude the possibility of penetration or overlap between the femur and the tibia, it was stipulated that only a single tangent to both the femoral and the tibial profiles exist at the contact point. Consequently, the normals to the femoral and the tibial profiles are always collinear, and their cross product must vanish. We thus get:

$$\mathbf{n}_1 \times \mathbf{n}_2 = \hat{0} \tag{8}$$

where $\mathbf{n}_1$ and $\mathbf{n}_2$ are the unit normal vectors to the femoral and tibial profiles, respectively, expressed in the femoral coordinate system. Equation (8) is rewritten in an expanded form as:

$$-R_{11}\frac{df_2(x')}{dx'} + R_{12} - R_{21}\frac{df_1(x)}{dx}\frac{df_2(x')}{dx'} + R_{22}\frac{df_1(x)}{dx} = 0.0; \ @\ x = x_c,\ x' = x_c' \tag{9}$$

Equation (9) represents the geometric compatibility condition.

Ligamentous Forces. The knee joint ligaments and the capsular tissue posterior to it were approximated by 10 nonlinear springs. Two springs represented the anterior and the posterior fibers of each of the anterior and the posterior cruciate ligaments, four springs represented the medial collateral ligament, and one spring represented each of the lateral collateral ligament and the posterior part of the capsule.

Let us define $\mathbf{S}_j$ as a vector parallel to the jth ligament and directed from the tibial insertion to the femoral origin, as shown in Fig. 2(*b*). This vector can be expressed as:

$$\mathbf{S}_j = \mathbf{R}_j - \mathbf{R}_0 - [T]\mathbf{R}_j' \tag{10}$$

where $\mathbf{R}_j$ and $\mathbf{R}_j'$ are the position vectors of the femoral and tibial insertion points of the jth ligaments, respectively. The unit vector in the direction of the jth ligament, λ_j, is written as $\lambda_j = \mathbf{S}_j/L_j$ where L_j is the length of the jth ligament. The force in the jth ligament is completely defined by $\mathbf{F}_j = F_j\lambda_j$ where F_j is the magnitude of the force in the jth ligament. The ligaments were assumed to carry load only when they were in tension, that is when the ligament length was larger than its slack length L_0. The magnitude of the force in the jth ligament was calculated using a quadratic force-elongation relation given by:

$$F_j = k_j(L_j - L_{0j})^2 \text{ if } L_j \geq L_{0j}; \text{ and } F_j = 0 \text{ if } L_j \leq L_{0j} \tag{11}$$

where k_j is the jth ligament stiffness coefficient. The values of these coefficients were obtained from Moeinzadeh et al. [7–11], and are listed in Table 1. The slack length of each ligament was obtained by assuming an extension ratio ϵ_j at full extension and using the following relation:

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Table 1 Ligament stiffness

Ligament	k (N/mm^2)
ACL, ant. fibers	20.0
ACL, post. fibers	10.0
PCL, ant. fibers	17.5
PCL, post. fibers	17.5
MCL, post. fibers	3.0
MCL, ant. fibers	6.0
MCL, oblique	3.0
MCL, deep	3.0
LCL	15.0
Posterior capsule	35.0

ACL: Anterior Cruciate Ligament.
PCL: Posterior Cruciate Ligament.
MCL: Medial Collateral Ligament.
LCL: Lateral Collateral Ligament.

(Force = $k\delta^2$; where δ is elongation in mm.)

Table 2 Extension ratios of the ligaments at full extension

Ligament	ϵ
ACL, ant. fibers	1.031
ACL, post. fibers	1.012
PCL, ant. fibers	1.050
PCL, post. fibers	1.050
MCL, post. fibers	1.050
MCL, ant. fibers	0.996
MCL, oblique	1.031
MCL, deep	1.049
LCL	1.050
Posterior capsule	1.050

ACL: Anterior Cruciate Ligament.
PCL: Posterior Cruciate Ligament.
MCL: Medial Collateral Ligament.
LCL: Lateral Collateral Ligament.

Table 3 Local coordinates of femoral and tibial insertion sites

Ligament	Femur x (mm)	Femur y (mm)	Tibia x' (mm)	Tibia y' (mm)
ACL, ant. fibers	−13.0	−17.0	−5.0	213.0
ACL, post. fibers	−9.0	−17.0	−2.0	213.0
PCL, ant. fibers	−2.0	−12.0	25.0	208.0
PCL, post. fibers	−22.0	−12.0	25.0	208.0
MCL, post. fibers	−13.0	−22.0	8.0	163.0
MCL, ant. fibers	2.0	−22.0	−7.0	173.0
MCL, oblique	−8.0	−22.0	30.0	203.0
MCL, deep	−5.0	−19.0	0.0	203.0
LCL	−15.0	−17.0	25.0	178.0
Posterior capsule	−30.0	−22.0	25.0	183.0

ACL: Anterior Cruciate Ligament.
PCL: Posterior Cruciate Ligament.
MCL: Medial Collateral Ligament.
LCL: Lateral Collateral Ligament.

$$\epsilon_j = \frac{L_j}{L_{0j}} \tag{12}$$

All extension values were obtained from Grood and Hefzy [17], and are listed in Table 2.

Figure 3 shows the different ligamentous structures at full extension where the coordinates of their femoral and tibial attachment points were obtained from Crowninshield et al. [18] and are listed in Table 3. Note that these coordinates are expressed with respect to the local body systems of axes. The local y coordinates of the ligamentous tibial attachments have large magnitudes since the local tibial coordinate system is taken at the center of mass of the tibia.

Contact Force. In this analysis, friction forces at the articulating surface were assumed negligible [19]. Due to the contact between the femur and the tibia, the femur exerts a force on the tibia as it moves along its articulating surface. The contact force, **N**, is applied normal to the tibial surface at the point of contact at $x = x_c$, and is expressed as $\mathbf{N} = N\mathbf{n}_1$ where N is the magnitude of the contact force.

External Forces. External forces include the weight and an applied load in the form of a forcing pulse. The weight of the leg, **W**, is applied on the tibia at its center of gravity. This force has a constant magnitude and a constant direction and is given by $\mathbf{W} = W\hat{\mathbf{j}}$. In this analysis, the weight of the leg was obtained from Moeinzadeh et al. [7-11] and was taken as $W = 30.868$ N which is equivalent to a mass of 3.15 kg.

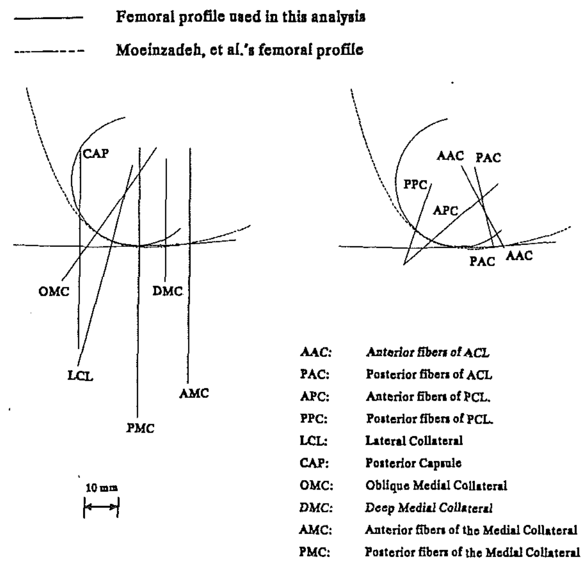

Fig. 3 *This figure shows the different ligamentous structures, and their tibial and femoral attachment locations at full extension. This figure also shows the tibial and femoral geometry used in this analysis. This figure is drawn to scale.*

The forcing pulse, **Fe**, is applied to the tibia at its center of gravity in a direction parallel to the x'-axis, and is given in a vector form as $\mathbf{Fe} = Fe\hat{\mathbf{i}}'$ where Fe is the magnitude of the force pulse. This force is transformed to the femoral system according to the following transformation:

$$\mathbf{FR} = (FR)_x\hat{\mathbf{i}} + (FR)_y\hat{\mathbf{j}} = R_{11}Fe\hat{\mathbf{i}} + R_{21}Fe\hat{\mathbf{j}} \tag{13}$$

where R_{11} and R_{12} are components of the rotational transformation matrix and are given by Eq. 1(c).

Equations of Motion. The tibia was assumed to undergo a general planar motion with respect to the fixed femur. The equations governing this motion, written with respect to the fixed femoral system of axes, are

$$(FR)_x + N(n_1)_x + \sum_{j=1}^{10} F_j(\lambda_j)_x = M\ddot{x}_0 \tag{14a}$$

$$(FR)_y + N(n_1)_y + \sum_{j=1}^{10} F_j(\lambda_j)_y = M\ddot{y}_0 \tag{14b}$$

$$([T]\mathbf{R}_c') \times (N\mathbf{n}_1) + \sum_{j=1}^{10} ([T]\mathbf{R}_j') \times (F_j\lambda_j) = I_z\ddot{\alpha} \tag{14c}$$

where M is the mass of the leg, $\ddot{x}_0$ and $\ddot{y}_0$ are the x and y components of the linear acceleration of the leg, I_z is the moment of inertia of the leg and foot around an axis perpendicular to the plane of motion and passing through the origin of the tibial system of the axes, and $\ddot{\alpha}$ is the angular acceleration of the leg and foot. In this analysis, the moment of inertia I_z was obtained from Moeinzadeh et al. [7-11] and was taken as 0.05 kg m^4.

Mathematical Solution

Knee joint motions are described by three differential equations, the equations of motion given by Eqs. 14(a)-14(c), and three nonlinear algebraic equations defining two contact conditions [Eqs. (6) and (7)] and a geometric compatibility condition [Eq. (9)]. This is a system of six equations in six

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unknowns; x_0 and y_0, the coordinates of the origin of the tibial system of axes with respect to the femoral system of axes, α, the orientation of the tibial system with respect to the femoral system, x_c and x_c', the x coordinates of the contact point in the femoral and the tibial systems of axes, respectively, and N the magnitude of the contact force. To solve this system, the Newmark method [13] is first used to reduce the differential equations to algebraic equations. These equations are then combined with the constraint equations to form a system of nonlinear algebraic equations. This system was solved using an iterative technique, namely the Newton-Raphson method [13].

Newmark Method. The analysis included dividing the time span of the motion into small time steps Δt and assuming the acceleration to be constant over each step. The acceleration at time station $t+\Delta t$ is ${}^{t+\Delta t}\ddot{U}$, and is written in the form [13]:

$$ {}^{t+\Delta t}\ddot{U}=\frac{4}{\Delta t^2}\,{}^{t+\Delta t}U-R_U \tag{15a}$$

where ${}^{t}\ddot{U}$ is the acceleration a t and R_U is a known quantity at time station $t+\Delta t$ and is given by:

$$ R_U=\frac{4}{\Delta t^2}\,{}^{t}U+\frac{4}{\Delta t}\,{}^{t}\dot{U}+{}^{t}\ddot{U} \tag{15b}$$

where ${}^{t}U$ is the displacement at station t, ${}^{t+\Delta t}U$ is the displacement at station $t+\Delta t$, ${}^{t}\dot{U}$ is the velocity at station t, and ${}^{t+\Delta t}\dot{U}$ is the velocity a $t+\Delta t$. Applying Eq. (15) to the x and y components of the linear acceleration and to the angular acceleration, and substituting into Eqs. (14) transforms the differential equations of motion into nonlinear algebraic equations. Using Eq. (15), one can find the acceleration at time station $t+\Delta t$ in terms of the position at stations t and $t+\Delta t$, and the velocity and acceleration at station t. Then, the velocity at time station $t+\Delta t$ can be found in terms of the velocity and acceleration at station t, and the acceleration at station $t+\Delta t$, as follows:

$$ {}^{t+\Delta t}\dot{U}={}^{t}\dot{U}+\frac{\Delta t}{2}\left({}^{t}\ddot{U}+{}^{t+\Delta t}\ddot{U}\right) \tag{16}$$

This numerical analysis will be employed to find the position, acceleration, and velocity at time $t+\Delta t$ in terms of those values at time t. Accordingly, these values at time $t = 0$ must be assumed to initiate the scheme. Since the tibia begins its motion from rest, the components of the velocity are equal to zero at time $t = 0$. The initial values of the x coordinates of the contact point in both femoral and tibial systems of axes were taken: ($x_c = -0.01326$ m) and ($x_c' = 0.00825$ m). The initial coordinates of the tibial origin with respect to the femur and the initial orientation of the tibia were calculated in terms of the initial values of x_c and x_c' using Eqs. (6), (7), and (9). Finally, the initial value of the contact force was assumed to satisfy equilibrium at time $t = 0$.

Newton-Raphson Method. At each time station $t+\Delta t$ combining Eqs. (14) with Eqs. (6), (7), and (9) produce a system of six nonlinear algebraic equations in six unknowns. To solve this system of equations the Newton-Raphson iteration technique was used. The solution to this nonlinear system amounts to finding a vector $\mathbf{U}^*$ which satisfies the equations:

$$ \mathbf{F}(\mathbf{U}^*)=0 \tag{17}$$

Assuming that in the iterative solution ${}^{t+\Delta t}\mathbf{U}^{(i-1)}$ have been evaluated at iteration $i-1$, the Newton-Rasphson solution of the system of equations given by Eq. (17) is thus written as:

$$ {}^{t+\Delta t}\mathbf{K}^{(i-1)}\Delta\mathbf{U}^{(i)}=-\,{}^{t+\Delta t}\mathbf{F}^{(i-1)} \tag{18a}$$

with

$$ \frac{\partial\mathbf{F}}{\partial\mathbf{U}}\Big|_{t+\Delta t U(i-1)}=-\,{}^{t+\Delta t}\mathbf{K}^{(i-1)} \tag{18b}$$

where ${}^{t+\Delta t}\mathbf{K}^{(i-1)}$ is the stiffness matrix and ${}^{t+\Delta t}\mathbf{F}^{(i-1)}$ is the residual, namely the load vector in iteration $i-1$, and $\Delta\mathbf{U}^{(i)}$ is the displacement increment which rather than producing an exact displacement, produces the next approximation given by:

$$ {}^{t+\Delta t}\mathbf{U}^{(i)}={}^{t+\Delta t}\mathbf{U}^{(i-1)}+\Delta\mathbf{U}^{(i)} \tag{19}$$

The criteria used in this analysis was

$$ |\Delta U_j^{(i)}|\le 0.0001 \tag{20}$$

where $\Delta U_j^{(i)}$ are the components of the vector $\Delta\mathbf{U}^{(i)}$.

If the solution has converged for all the elements of vector $\Delta\mathbf{U}^{(i)}$, the displacement vector, the load vector, and the stiffness matrix values at this time station are set as initial values for the next station, otherwise the iteration is continued.

In this analysis, a time step Δt of 0.0001 s was used. A larger time step affected the accuracy of the solution, and caused the Newton-Raphson iteration to diverge. A smaller value of Δt affected the stability of the system because it increased the ill-conditioning number of the stiffness matrix [13]. It was found that $\Delta t = 0.0001$ s represents the best trade off between accuracy and stability for the problem under consideration.

Equation 18(b) indicates that the stiffness matrix is the partial differential of each component of the load vector with respect to each of the independent variables of the system. Expressions for the load vector are determined from Eq. (17) as:

$$ F_1=-x_c+x_0+R_{11}x_c'+R_{12}f_2(x_c') \tag{21}$$

$$ F_2=-f_1(x_c)+y_0+R_{21}x_c'+R_{22}f_2(x_c') \tag{22}$$

$$ F_3=-R_{11}\frac{df_2(x')}{dx'}+R_{12}-R_{21}\frac{df_1(x)}{dx}\frac{df_2(x')}{dx'}+R_{22}\frac{df_1(x)}{dx};\ @\ x=x_c\ x'=x_c' \tag{23}$$

$$ F_4=-(FR)_x-N(n_1)-\sum_{j=1}^{10}F_j(\lambda_j)_x+M\left(\frac{4}{\Delta t^2}\,{}^{t+\Delta t}x-R_x\right) \tag{24}$$

$$ F_5=-(FR)_y-N(n_1)_y-\sum_{j=1}^{10}F_j(\lambda_j)_y+M\left(\frac{4}{\Delta t^2}\,{}^{t+\Delta t}y-R_y\right) \tag{25}$$

$$ F_6=-([T]\mathbf{R}_c')\times(N\mathbf{n}_1)-\sum_{j=1}^{10}([T]\mathbf{R}_j'\times(F_j\lambda_j)+I_z\left(\frac{4}{\Delta t^2}\,{}^{t+\Delta t}\alpha-R_\alpha\right) \tag{26}$$

Results

In what follows, we present results for knee motions under a sudden impact simulated by a posterior forcing pulse in the form of a rectangular step function defined as:

$$ F_e=A[H(t)-H(t-t_o)] \tag{27}$$

where A is the pulse amplitude, t_o is the pulse duration, and $H(t)$ is a step function defined as:

$$ H(t-t_o)=1\quad t\ge t_o \tag{28a}$$

$$ H(t-t_o)=0\quad t<t_o \tag{28b}$$

The posterior forcing pulses are applied to the tibia when the knee joint is at full extension; knee motions are tracked until 90 degrees of knee flexion. Results are presented to show the effects of varying the pulse amplitude and duration on the magnitude and location of the contact force and on the different ligamentous forces.

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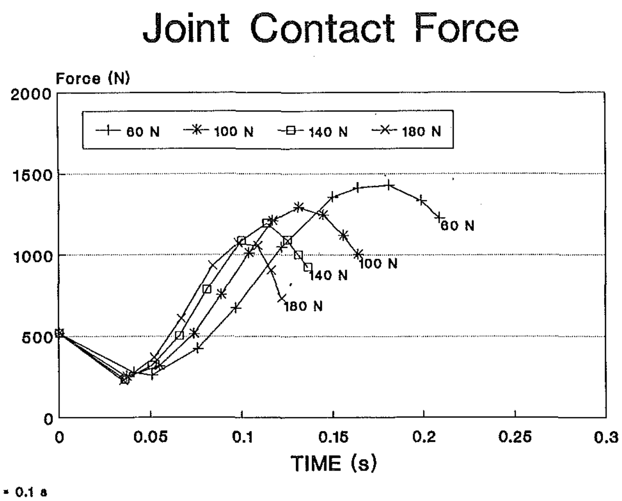

Fig. 4 This figure shows how the joint contact force changes with time for different rectangular forcing pulses having a constant pulse duration of 0.1 s and different pulse amplitudes

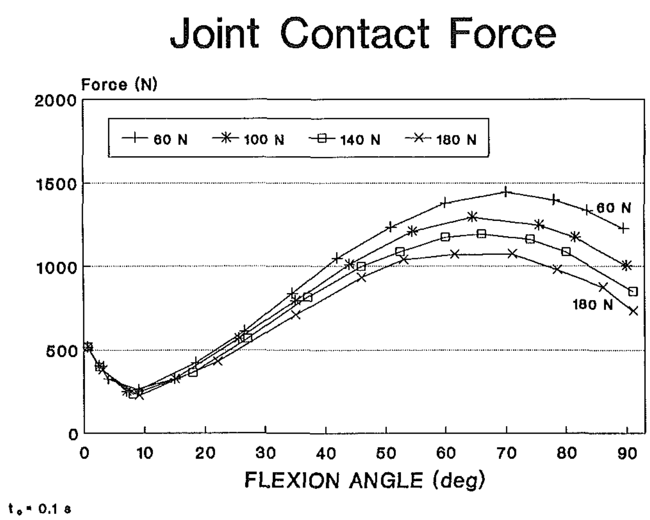

Fig. 5 This figure shows how the joint contact force changes with knee flexion for different rectangular forcing pulses having a constant pulse duration of 0.1 s and different pulse amplitudes

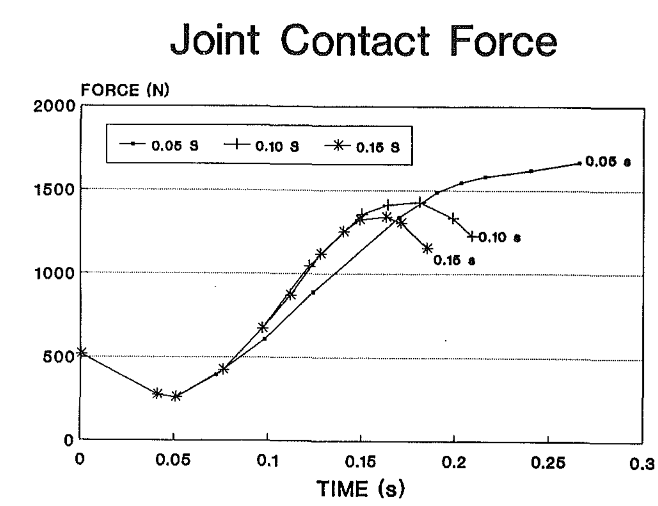

Fig. 6 This figure shows how the joint contact force changes with time for different rectangular forcing pulses having a constant pulse amplitude of 60.0 N and different pulse durations

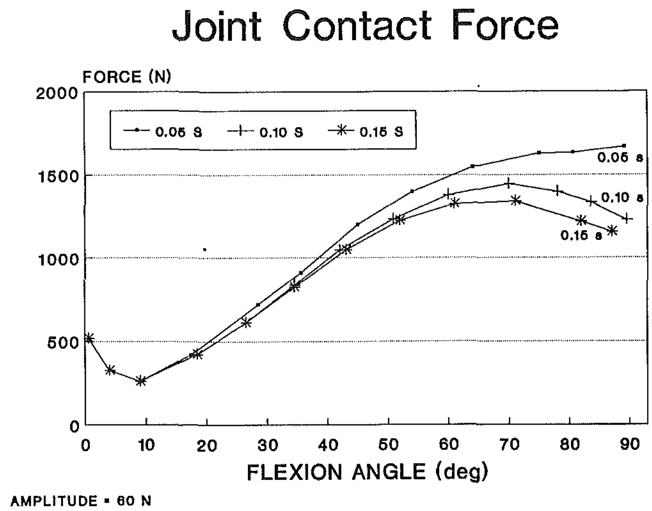

Fig. 7 This figure shows how the joint contact force changes with knee flexion for different rectangular forcing pulses having a constant pulse duration of 0.1 s and different pulse amplitudes

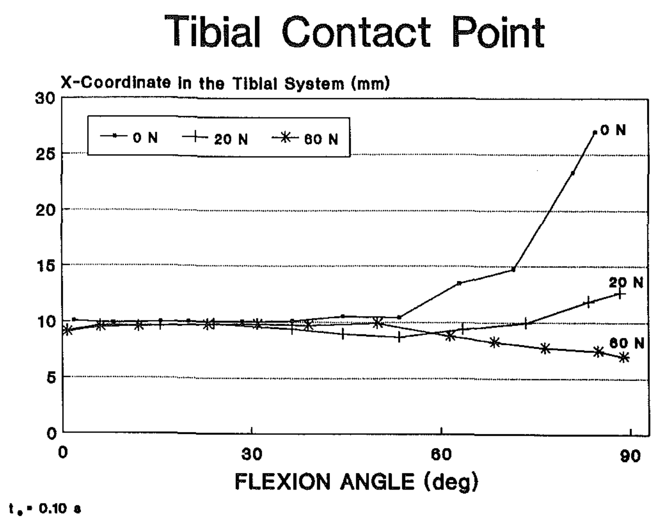

Fig. 8 This figure shows how the tibial x-coordinate of the contact point changes with knee flexion for different rectangular forcing pulses having a constant pulse duration of 0.10 s and pulse amplitudes of 0.0 N, 20.0 N and 60.0 N

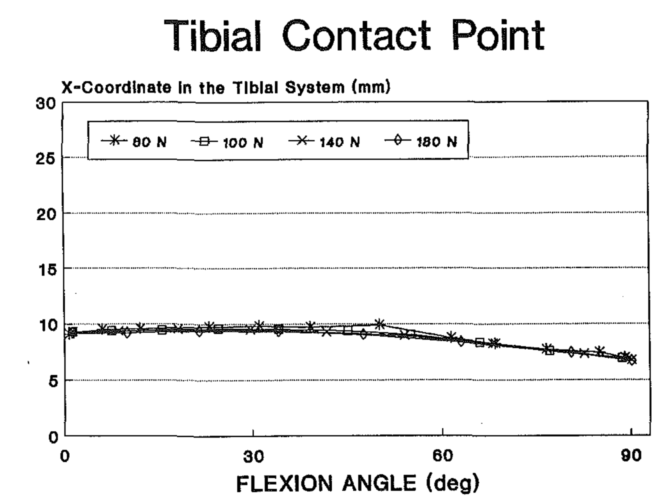

Fig. 9 This figure shows how the tibial x-coordinate of the contact point changes with knee flexion for different rectangular forcing pulses having a constant pulse duration of 0.10 s and pulse amplitudes of 60.0 N, 100.0 N, 140.0 N and 180.0 N

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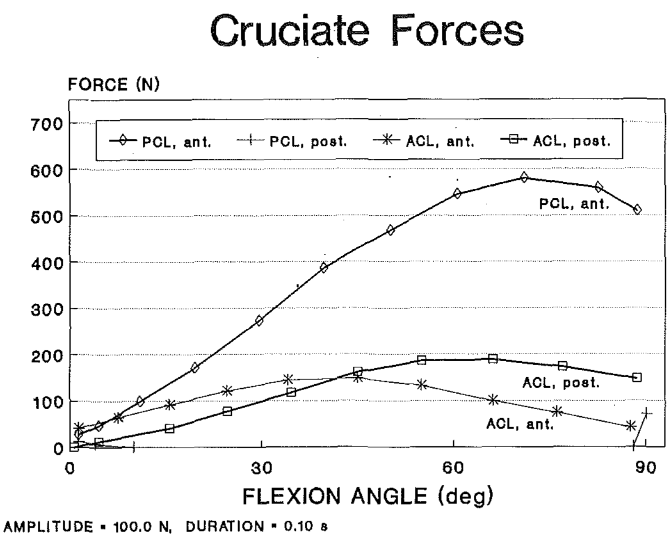

Fig. 10 This figure shows how the forces in the anterior and posterior fibers of the anterior and posterior cruciate ligaments change with knee flexion for a rectangular forcing pulse having a pulse amplitude of 100.0 N and a pulse duration of 0.1 s

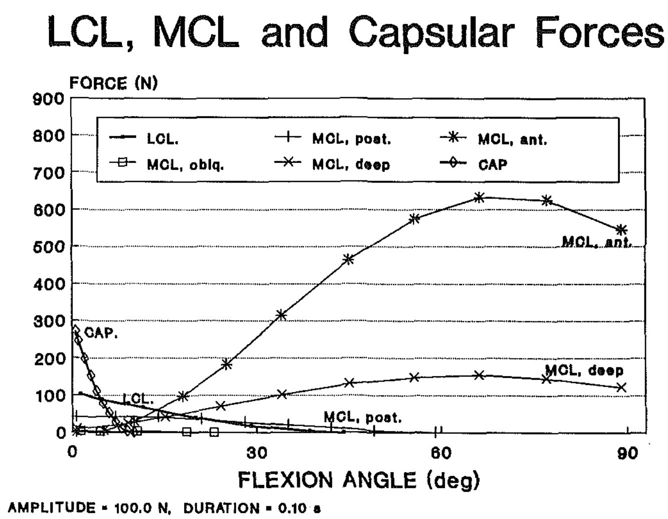

Fig. 11 This figure shows how the forces in the lateral collateral, the posterior capsule and the posterior, anterior, oblique and deep fibers of the medial collateral ligament change with knee flexion for a rectangular forcing pulse having a pulse amplitude of 100.0 N and a pulse duration of 0.1 s

(1) Magnitude of the Tibio-Femoral Contact Force. Figure 4 shows that increasing the pulse amplitude produced a decrease in the motion duration from full extension to full flexion. Figure 5 shows that the contact force decreased in the first 10 degrees of flexion, then increased until the knee reached about 65 degrees of flexion, and then decreased until 90 degrees of knee flexion. The results indicate that the maximum contact force decreased as the amplitude of the forcing pulse increased.

Figure 6 shows that increasing the pulse durations produced a decrease in the time it takes the knee to reach 90 degrees of knee flexion. Figure 7 shows that the magnitude of the contact force decreased in the first 10 degrees of flexion, then increased until about 65 degrees of flexion, and then decreased until 90 degrees of knee flexion. The results indicate that the maximum contact force decreased as the duration of the forcing pulse increased.

(2) Location of the Tibio-Femoral Contact Force. The results shown in Fig. 8 indicate that the tibial contact point moved posteriorly 17 mm (millimeters) as the knee was flexed from 0 to 85 degrees of knee flexion under its own weight and when no posterior pulsing force was applied to the tibia (positive x is posterior on the tibia). Increasing the posterior pulsing force to 20.0 N caused the posterior motion of the tibial contact point to decrease to 3 mm as the knee was flexed from 0 to 90 degrees. When the posterior pulsing force increased to 60 N, this motion was reversed and the tibial contact point moved 3 mm anteriorly as the knee was flexed from 0 to 90 degrees of flexion. The results shown in Fig. 9 show that this motion was not altered as the posterior pulsing force increased from 60.0 N to 180.0 N.

It was also found that the femoral contact point moved posteriorly as the knee was flexed from 0 to 90 degrees of knee flexion. The motion pattern of the femoral contact point was independent of pulse amplitude or pulse duration.

(3) Ligamentous Forces. The results shown in Fig. 10 indicate that the posterior fibers of the anterior cruciate ligament sustained more tension than the anterior fibers of that ligament in the range of 45 to 90 degrees of knee flexion; this trend was reversed in the first 45 degrees of knee flexion. The posterior fibers of the posterior cruciate ligament carried tension only in the first 10 degrees of knee flexion, and in the last 5 degrees of flexion: from 85 to 90 degrees. The anterior fibers of the posterior cruciate ligament carried relatively large forces at all flexion angles, with a maximum value of about 580.0 N.

Figure 11 shows that the lateral collateral ligament sustained forces only in the first 45 degrees of knee flexion, with a maximum value of 100.0 N occurring at full extension. This figure also shows that the posterior capsule sustained forces only in the first 10 degrees of knee flexion, with a maximum of 290.0 N occurring at full extension. Figure 11 also shows that the oblique fibers of the medial collateral ligament carried very small forces. Within the medial collateral ligament, the anterior fibers carried more forces than the deep fibers which carried more forces than the posterior fibers in the range of 15 to 90 degrees of knee flexion. The maximum force in the anterior, deep, and posterior fibers of this ligament in the range of 15 to 90 degrees of knee flexion averaged 620.0 N, 160.0 N, and 50.0 N, respectively. This trend was reversed in the first 15 degrees of knee flexion where the maximum force in all these fiber bundles of the medial collateral ligament reached 50.0 N at 15 degrees of knee flexion. Finally, the maximum force in the anterior and deep fibers occurred from 70 to 80 degrees of knee flexion, while the maximum force in the posterior fibers occurred at 15 degrees of knee flexion.

Discussion

This work presents a two-dimensional dynamic model of the human knee joint that predicts its response under sudden impact loads. This analysis uses a differential form of the Newton-Raphson method differentiating the system equations with respect to the six independent variables and six equations in six unknowns are then solved. In the formulation presented here all the coordinates of the ligamentous attachment points were dependent variables, as a result introducing more ligaments or splitting each ligament to several fiber bundles does not affect the system to be solved. Also, in these calculations, a femoral profile [15], that allowed us to predict knee response over a range of motion from 0 to 90 degrees of knee flexion is used.

It is hard to compare this model calculation with other data because of the limited amount of experimental data available in the literature for the dynamic behavior of the human knee joint. As reported by Dortmans et al. [20], the dynamic behavior of the joint must be described in terms of the loads exerted on the joint, the six components of the three-dimensional motion of the tibia with respect to the femur, and the deformations of the different components forming the joint and including the ligaments, meniscii, and cartilage. Dortmans et al. [20] stated that, at this stage, "... the dynamic behavior

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described above is much too complicated to deal with, due to a lack of experimental techniques to quantify all parameters."

The results obtained from our model are different in nature from those reported by Jans et al. [21] and Dortmans et al. [20, 22] since they presented results in terms of transfer functions between the applied load and the displacements of the tibia. Yet, our model calculations are indirectly in agreement with Dortmans et al. [20, 22] data which suggest that increasing load level causes a decrease in the joint stiffness. Our results show that the contact force decreased as the load level increased, which is an indirect indication of a decrease in the joint stiffness.

It was found that the femoral contact point always moved posteriorly with knee flexion. This is in agreement with the results reported by Moeinzadeh et al. [17-11]. The results shown in Fig. 8 indicate that the tibial contact point moved posteriorly as the knee was flexed from 0 to 85 degrees of knee flexion under its own weight and when no posterior pulsing force was applied to the tibia. These results are in agreement with those reported by Muller [23]. Increasing the posterior pulsing force to 20.0 N caused the posterior motion of the tibial contact point to decrease. When the posterior pulsing force increased to 60 N, this motion was reversed and the tibial contact point moved anteriorly as the knee was flexed from 0 to 90 degrees of flexion. This motion was not altered as the posterior pulsing force increased from 60.0 N to 180.0 N.

The results describing the motion of the tibial contact point are expected since they correspond to a dynamic flexion of the tibia due to a posterior forcing pulse. This motion can be thought of as the resultant of two motions: flexion of the tibia which causes the contact point to move posteriorly on the tibia and posterior sag of the tibia which causes the contact point to move anteriorly on the tibia.

Differences of opinion still exist with regard to the knee ligaments' function; this has been noted by Edwards et al. [24], France et al. [25], Ahmed et al. [26] and Markolf et al. [27]. There are several and different reasons for these differences, depending on whether the ligamentous forces are being determined experimentally or being predicted using a mathematical model.

Model predictions depend on the values of the stiffness of the ligaments. Yet, a considerable conflict exists in the literature regarding these data. Furthermore, it has been recently reported by Figgie et al. [28], Butler et al. [29], and Rogers et al. [30] that the ligaments have different strengths depending on the flexion angle and the orientation of the ligament at which the test is done. In these calculations, and as recommended by Butler et al. [3], stiffness values from the higher range of those reported in literature were used but the change of the ligamentous stiffnesses due to the change of flexion angle was not accounted.

Experimental data available in the literature, describing knee ligament forces as function of knee motions, are confined to static situations [24-27]. Therefore, a detailed comparison of the results of the present dynamic model with literature data does not seem valid; only comparable trends could be reported. Keeping this in mind, it is found that the results obtained from this model show a general agreement with the quasistatic experimental investigations reported in the literature and related to the function of the ligamentous structures. For instance, the results indicate that forces in the anterior fibers of the posterior cruciate ligament were maximum around 75 degrees of knee flexion. This is in agreement with France et al. [25] who reported that the anterior component of the posterior cruciate ligament was maximally taut at 80 degrees of knee flexion. The present results for the anterior cruciate ligament are not in agreement with what is reported in the literature. This is a result of the two-dimensional nature of the present model. Confined to the sagittal plane, this model does not really distinguish or differentiate fully between the anterior and posterior fibers of the anterior cruciate ligament which run parallel to each other in a sagittal view but cross each other in a frontal view [18].

The force in the lateral collateral ligament was maximum at full extension, and decreased with knee flexion until it reached zero around 50 degrees of knee flexion. These results are in agreement with Moeinzadeh et al. [7-11], Crowninshield et al. [18], Edwards et al. [24], Wang et al. [32] and Ogata [33] who reported a maximal strain at full extension.

The posterior part of the knee capsule was found to be effective in resisting motion but only around full extension and that it was totally relaxed after 10 degrees of flexion which is in agreement with Crowninshield et al. [18] who reported that the posterior capsule performs a function only in the first few degrees of knee flexion.

The results show that the posterior capsule and the collaterals carry most of the load at full extension. This is to support the clinical observation, reported by Ogata et al. [33], that posterior sag does not occur at full extension in a PCL deficient knee unless the collateral ligaments were injured. Further posterior sag occurring in PCL deficient knees after sectioning the posterior capsule is a proof of the importance of the collaterals and the posterior capsule in stabilizing the knee at full extension.

Furthermore, we found that the major fiber bundles resisting the posterior force pulse in the range of 20 to 90 degrees of knee flexion are the anterior fibers of the posterior cruciate ligament and the medial collateral ligament explaining the clinical finding reported by Cross and Powell [34], Loos et al. [35], McMaster [36], and Trickey [37] that most of isolated posterior cruciate ligament injuries or combined injuries of the posterior cruciate and the medial collateral result from car dash board injuries, motor cycle accidents, or, a fall on a flexed knee which produce a posterior impact on the flexed knee causing these isolated or combined injuries.

Using the mathematical formulation described in this paper makes it easier to expand this work to three-dimensions in order to describe the general motion of the tibia with respect to the femur. This will include the development of a three-dimensional mathematical representation of the articular surfaces of the femur and tibia to be used in the model. Associated problems include the difficulties in the numerical evaluation of the stiffness matrix caused by having more than one mathematical representation of these surfaces. These difficulties occur at the common boundaries of the surfaces. Such a three-dimensional model is needed in order to simulate real knee response during sudden impact as it would result in car or sports injuries.

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